DEVIXO SOLUTIONS

Task 01: Python Programming, Data Analysis & Visualization

Project: Netflix Movies and TV Shows Data Analysis

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Internship Domain: AI/ML

Submission Date: 10 November 2026

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1. Introduction

Data analysis is an important process for understanding datasets, identifying patterns, and extracting meaningful information. This project focuses on analyzing the Netflix Movies and TV Shows dataset using Python libraries.

The project covers data loading, data cleaning, exploratory data analysis (EDA), statistical analysis, and data visualization. Different visualizations are used to explore Netflix content types, release years, ratings, countries, genres, and content durations.

2. Problem Statement

The purpose of this project is to analyze the Netflix Movies and TV Shows dataset and identify useful patterns and trends.

The dataset contains missing values and different types of information, including numerical and categorical features. The task involves cleaning the data, exploring its characteristics, performing statistical analysis, and presenting findings through visualizations.

3. Project Objectives

- Load and understand the Netflix dataset.
- Identify missing values and duplicate records.
- Clean and prepare the dataset for analysis.
- Perform exploratory data analysis.
- Calculate descriptive statistics.
- Create at least ten different visualizations.
- Identify meaningful insights from the dataset.
- Summarize the findings and conclusions.

4. Dataset Description

Dataset Name: Netflix Movies and TV Shows

Dataset Source: Kaggle

Dataset Link: https://www.kaggle.com/datasets/shivamb/netflix-shows

Dataset Size: 8,807 rows and 12 columns.

The dataset contains information about Netflix movies and television shows, including titles, directors, cast members, countries, release years, ratings, durations, genres, and descriptions.

4.1 Features / Columns

Column	Description
show_id	Unique content identifier
type	Movie or TV Show
title	Content title
director	Director name
cast	Cast members
country	Country of production
date_added	Date added to Netflix
release_year	Original release year
rating	Content rating
duration	Movie duration or TV show seasons
listed_in	Content genres
description	Content description

5. Tools and Technologies

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Kaggle Dataset

6. Methodology and Implementation

6.1 Data Loading

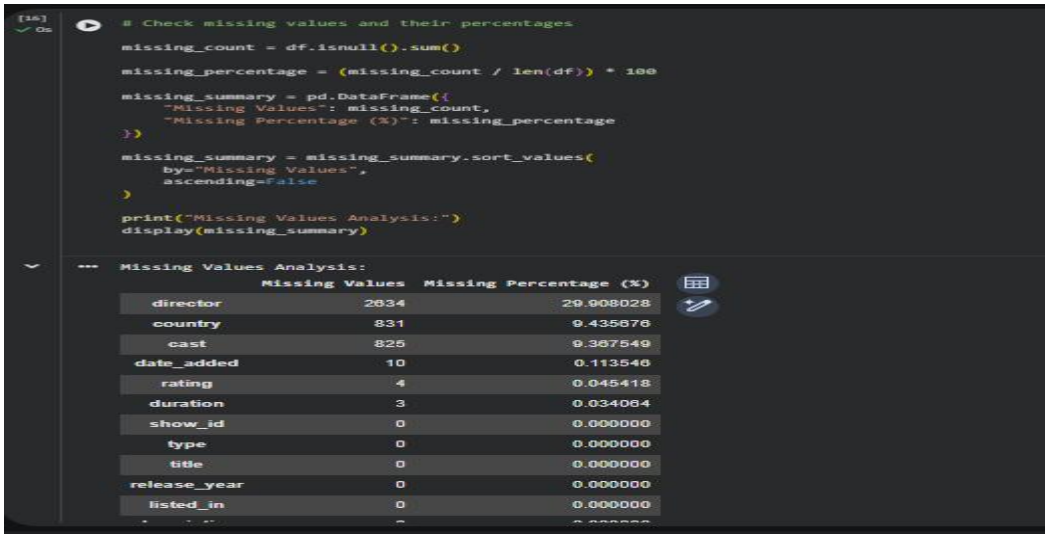
The required Python libraries were imported, and the Netflix dataset was loaded using Pandas.

The following operations were performed:

- Displayed the first five records.
- Displayed the last five records.
- Displayed a random sample.
- Checked dataset dimensions.
- Examined column names and data types.
- Identified numerical and categorical columns.
- Checked missing values.

Initial Dataset Shape: 8,807 rows × 12 columns.

Dataset Information and Missing Values:

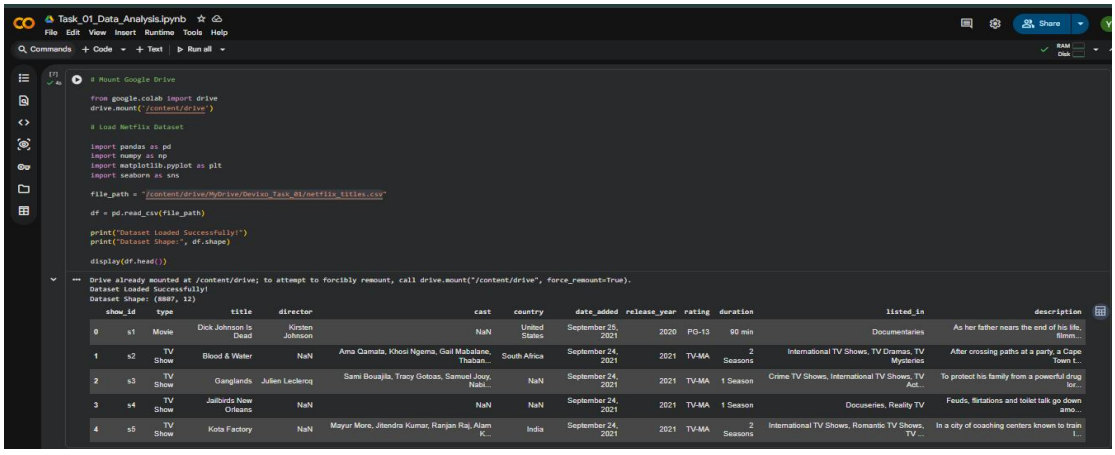


```
# Check missing values and their percentages
missing_count = df.isnull().sum()
missing_percentage = (missing_count / len(df)) * 100
missing_summary = pd.DataFrame({
    "Missing Values": missing_count,
    "Missing Percentage (%)": missing_percentage
})
missing_summary = missing_summary.sort_values(
    by="Missing Values",
    ascending=False
)
print("Missing Values Analysis:")
display(missing_summary)
```

```
Missing Values Analysis:
```

	Missing Values	Missing Percentage (%)
director	2634	29.908028
country	831	9.435676
cast	825	9.367549
date_added	10	0.113546
rating	4	0.045418
duration	3	0.034064
show_id	0	0.000000
type	0	0.000000
title	0	0.000000
release_year	0	0.000000
listed_in	0	0.000000

Dataset Loading and Preview:



```
# Mount Google Drive
from google.colab import drive
drive.mount('/content/drive')
# Load Netflix Dataset
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
file_path = "/content/drive/MyDrive/Devixo_Task_01/netflix_titles.csv"
df = pd.read_csv(file_path)
print("Dataset Loaded Successfully!")
print("Dataset Shape:", df.shape)
display(df.head())
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
Dataset Loaded Successfully!
Dataset Shape: (8807, 12)
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...

6.2 Data Cleaning

The dataset was examined for missing values, duplicate records, incorrect data types, and inconsistent values.

The following cleaning operations were performed:

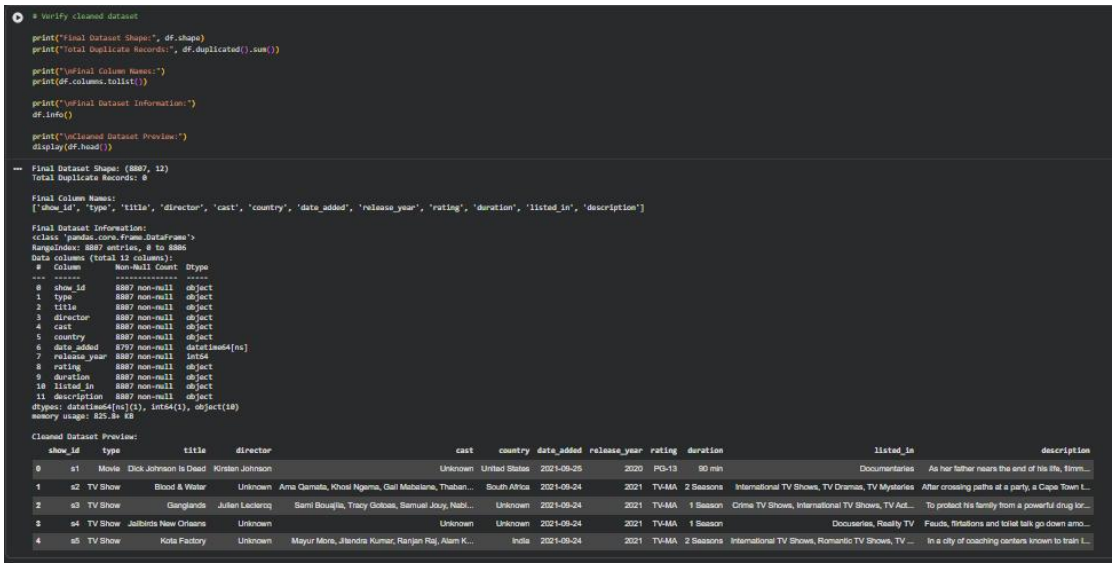
1. Identified missing values in director, cast, country, date_added, rating, and duration.
2. Checked duplicate records. No complete duplicate records were found.
3. Verified that all 8,807 show IDs were unique.
4. Filled missing director, cast, country, and rating values with "Unknown".
5. Converted the date_added column from object to datetime format.
6. Identified three duration values incorrectly stored in the rating column: 66 min, 74 min, and 84 min.
7. Moved these values into the duration column and corrected their rating values.
8. Retained the ten missing date_added values because their dates were unavailable.

Final Dataset Shape: 8,807 rows × 12 columns.

Remaining Missing Values: 10, all in date_added.

Duplicate Records: 0.

Final Cleaned Dataset :



```
# Verify cleaned dataset

print("Final Dataset Shape:", df.shape)
print("Total Duplicate Records:", df.duplicated().sum())

print("\nFinal Column Names:")
print(df.columns.tolist())

print("\nFinal Dataset Information:")
df.info()

print("\nCleaned Dataset Preview:")
display(df.head())
```

```
Final Dataset Shape: (8807, 12)
Total Duplicate Records: 0

Final Column Names:
['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added', 'release_year', 'rating', 'duration', 'listed_in', 'description']

Final Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      8807 non-null   object
 4   cast          8807 non-null   object
 5   country       8807 non-null   object
 6   date_added    8797 non-null   datetime64[ns]
 7   release_year  8807 non-null   int64
 8   rating        8807 non-null   object
 9   duration      8807 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: datetime64[ns](1), int64(1), object(10)
memory usage: 825.8+ KB

Cleaned Dataset Preview:
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	Unknown		Unknown	Unknown	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	Unknown	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...

6.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the dataset and identify patterns.

The analysis included:

- Distribution of Movies and TV Shows.
- Release year analysis.
- Content rating distribution.
- Country-wise content analysis.
- Genre distribution.

- Movie duration analysis.
- TV Show seasons analysis.
- Content additions by year and month.

Content Type Distribution

Content Type	Count	Percentage
Movies	6,131	69.62%
TV Shows	2,676	30.38%
Total	8,807	100%

Movies represent the majority of records in the dataset.

Release Year Analysis

- Earliest release year: 1925.
- Latest release year: 2021.
- Most common release year: 2018.
- Number of titles released in 2018: 1,147.

Content Rating Analysis

The most common content rating was TV-MA, with 3,207 entries, representing approximately 36.41% of the dataset.

Country Analysis

The top five country entries were:

Country	Count
United States	3,689
India	1,046
United Kingdom	804
Canada	445
France	393

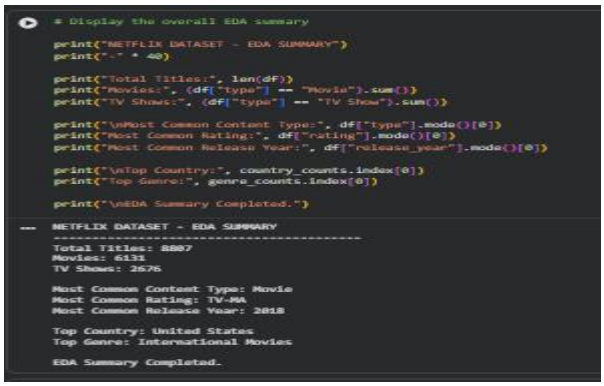
Genre Analysis

The five most frequent genre entries were:

Genre	Count
International Movies	2,752
Dramas	2,427
Comedies	1,674
International TV Shows	1,351
Documentaries	869

Country and genre counts include entries from titles associated with multiple countries or genres. Therefore, these counts should not be interpreted as unique title counts.

EDA Results:



```
# Display the overall EDA summary

print("NETFLIX DATASET - EDA SUMMARY")
print("-" * 40)

print("Total Titles:", len(df))
print("Movies:", (df["type"] == "Movie").sum())
print("TV Shows:", (df["type"] == "TV Show").sum())

print("\nMost Common Content Type:", df["type"].mode()[0])
print("Most Common Rating:", df["rating"].mode()[0])
print("Most Common Release Year:", df["release_year"].mode()[0])

print("\nTop Country:", country_counts.index[0])
print("Top Genre:", genre_counts.index[0])

print("\nEDA Summary Completed.")
```

```
NETFLIX DATASET - EDA SUMMARY
----------------------------------------
Total Titles: 8807
Movies: 6131
TV Shows: 2676

Most Common Content Type: Movie
Most Common Rating: TV-MA
Most Common Release Year: 2018

Top Country: United States
Top Genre: International Movies

EDA Summary Completed.
```

6.4 Statistical Analysis

Descriptive statistics were calculated to understand numerical features.

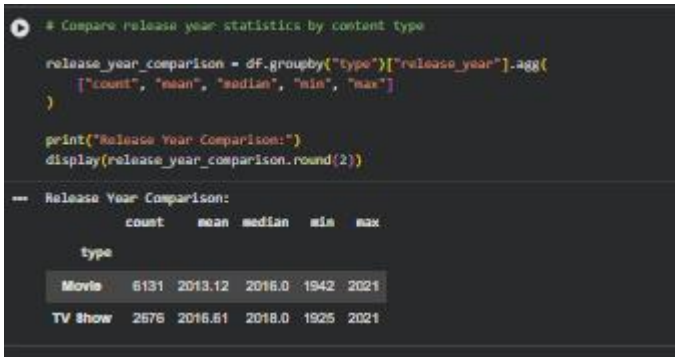
The analysis included mean, median, mode, minimum, maximum, range, standard deviation, and variance.

Important statistical findings:

Metric	Result
Average movie duration	99.56 minutes
Average TV Show duration	1.76 seasons
Minimum release year	1925
Maximum release year	2021
Median release year	2017
Mean	2014.18

Movie duration and TV Show seasons were analyzed separately because they use different units.

Statistical Analysis Output:



```
# Compare release year statistics by content type

release_year_comparison = df.groupby("type")["release_year"].agg(
    ["count", "mean", "median", "min", "max"]
)

print("Release Year Comparison:")
display(release_year_comparison.round(2))
```

```
Release Year Comparison:
```

type	count	mean	median	min	max
Movie	6131	2013.12	2016.0	1942	2021
TV Show	2676	2016.61	2018.0	1925	2021

7. Data Visualizations

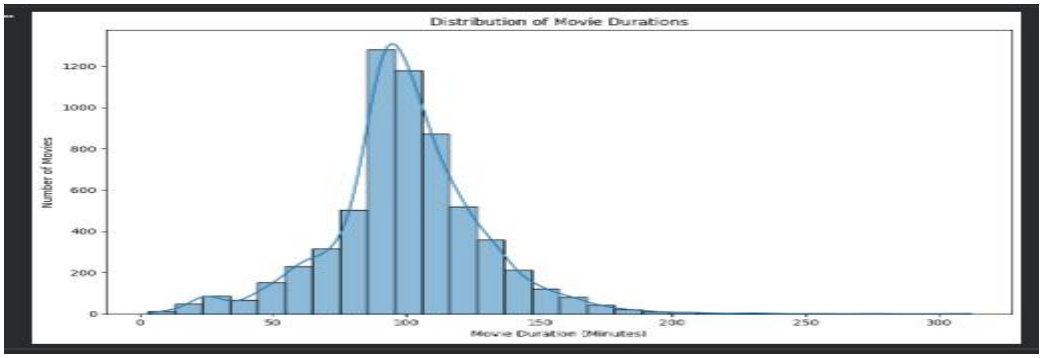
Ten different visualizations were created using Matplotlib and Seaborn.

No.	Visualization	Purpose
1	Histogram	Analyzes movie duration distribution
2	Bar Chart	Displays the top release years
3	Pie Chart	Compares Movies and TV Shows
4	Scatter Plot	Explores release year and movie duration
5	Line Chart	Shows yearly content additions
6	Box Plot	Examines movie duration distribution and outliers
7	Correlation Heatmap	Displays correlations between numerical features
8	Count Plot	Shows the number of Movies and TV Shows
9	Pair Plot	Explores relationships between numerical features
10	Violin Plot	Compares movie duration distributions across release periods

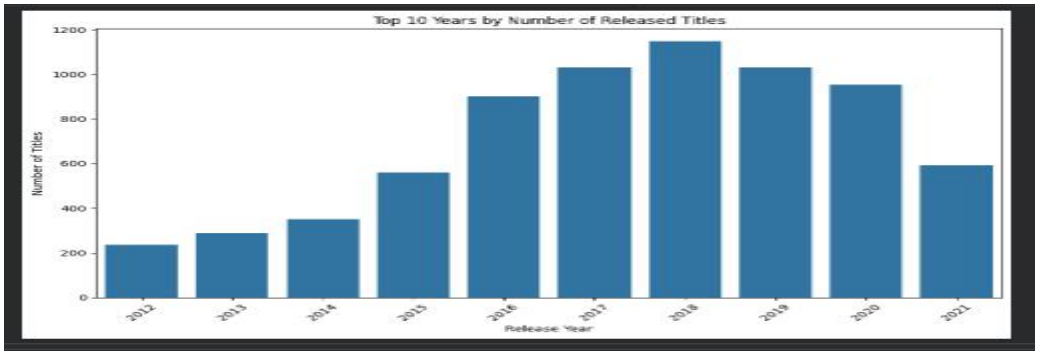
Each visualization includes an appropriate title, axis labels, and readable dimensions.

Note: Movie duration is measured in minutes, while TV Show duration is measured in seasons. These different units should not be directly compared through correlation.

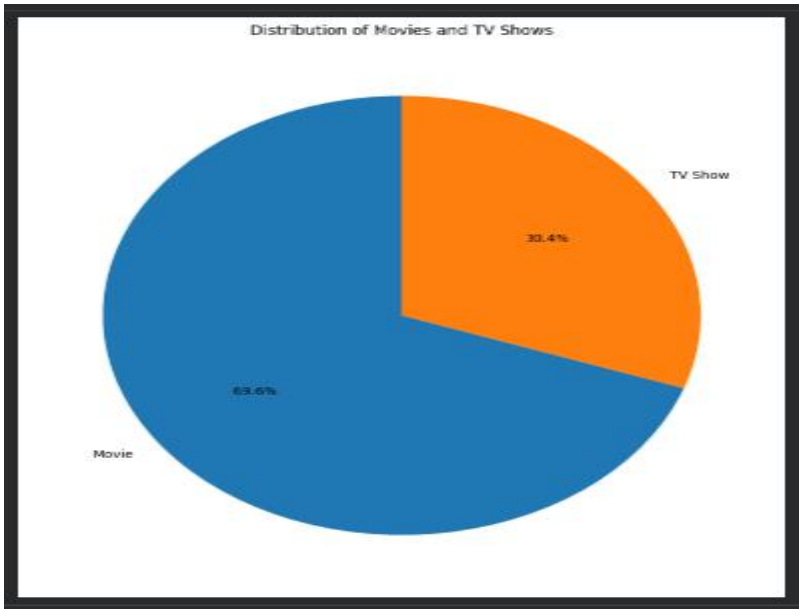
Histogram:



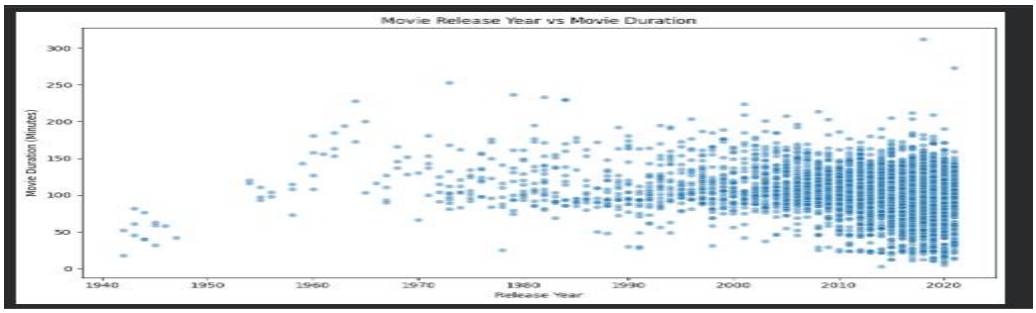
Bar Chart:



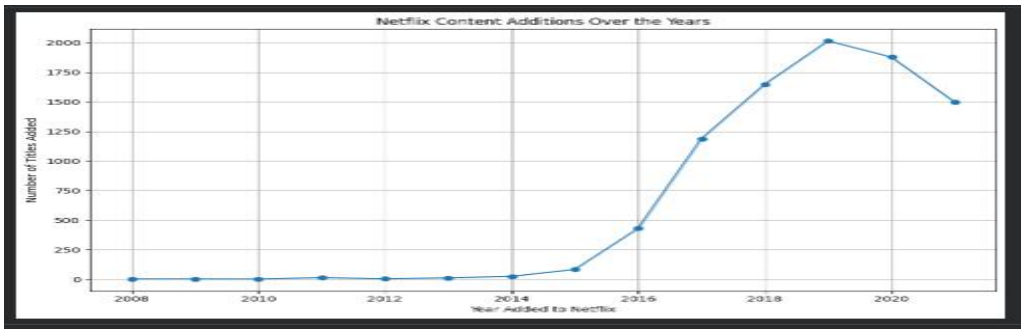
Pie Chart:



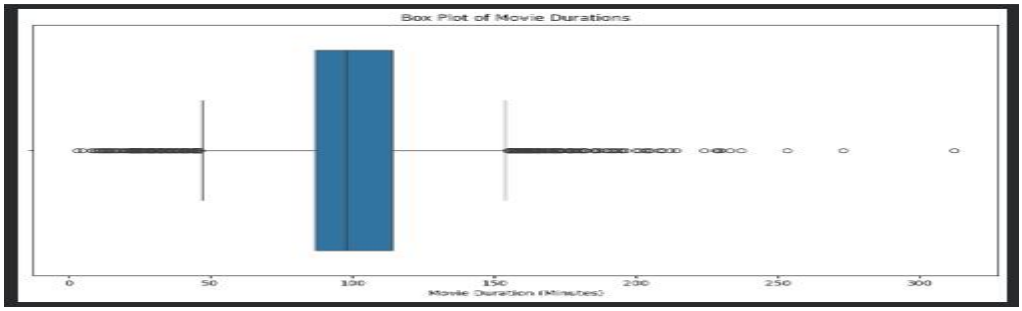
Scattar Plot:



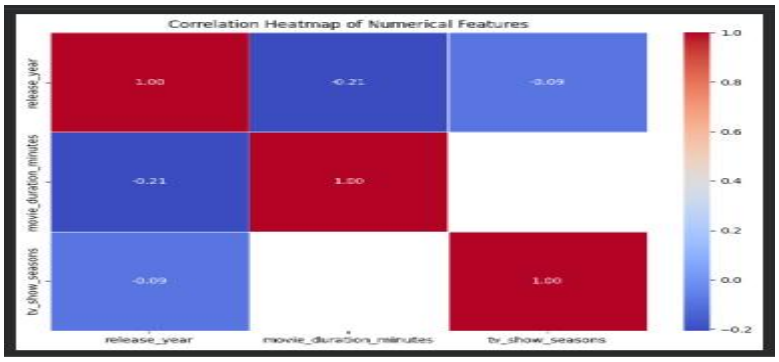
Line Chart:



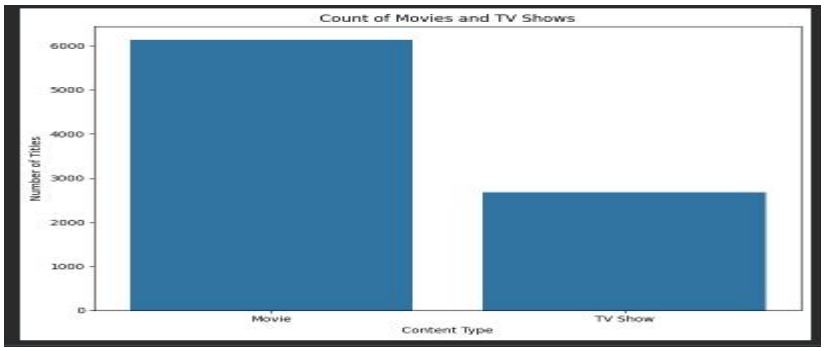
Box Plot:



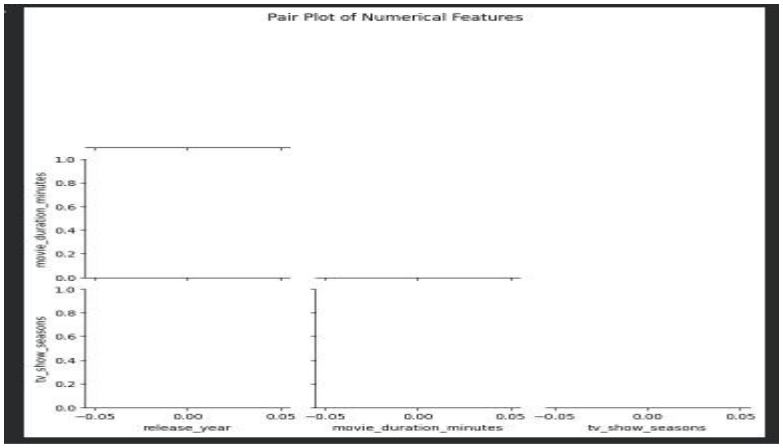
Correlation Heatmap:



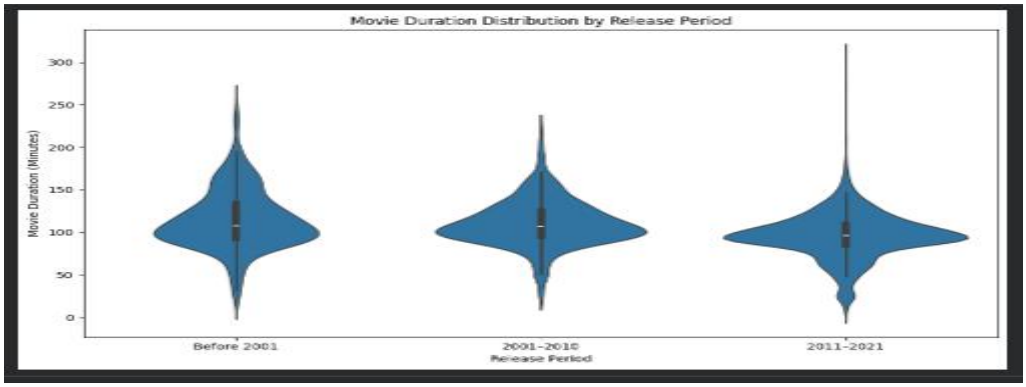
Count Plot:



Pair Plot:



Violin Plot:



8. Major Findings and Key Insights

Based on the analysis, the following insights were identified:

1. The dataset contains 8,807 titles.
2. Movies account for 69.62% of the dataset, while TV Shows account for 30.38%.
3. The content release years range from 1925 to 2021.
4. The year 2018 has the highest number of released titles, with 1,147 titles.
5. TV-MA is the most common rating, with 3,207 entries.
6. The United States has the highest country count, with 3,689 entries.
7. International Movies is the most frequent genre entry, with 2,752 entries.
8. Movies have an average duration of approximately 99.56 minutes.
9. TV Shows have an average duration of approximately 1.76 seasons.
10. Netflix added the highest number of titles in 2019, with 2,016 additions.

Additional Observation: July recorded the highest number of content additions, with 827 entries, while February recorded the lowest, with 563 entries

Key Insights Output:

```
--- 1. Total titles in the dataset: 8807

    2. Content type distribution:
    type
    Movie      6131
    TV Show    2676
    Name: count, dtype: int64

    3. Content type percentages:
    type
    Movie      69.62
    TV Show    30.38
    Name: proportion, dtype: float64

    4. Most common release year:
    2018

    5. Most common content rating:
    TV-MA

    6. Top 5 countries:
    country
    United States     3689
    India             1046
    United Kingdom     804
    Canada             445
    France             393
    Name: count, dtype: int64

    7. Top 5 genres:
    listed_in
    International Movies      2752
    Dramas                    2427
    Comedies                  1674
    International TV Shows    1351
    Documentaries              869
    Name: count, dtype: int64

    8. Average movie duration:
    99.56 minutes

    9. Average TV show seasons:
    1.76

    10. Year with the most Netflix content additions:
    2019.0 - 2016 titles

    11. Month with the most Netflix content additions:
    July - 827 titles
```

9. Challenges and Solutions

Challenge	Solution
Missing values in multiple columns	Filled missing metadata with "Unknown" where appropriate
Incorrect values in rating column	Corrected three misplaced duration values
Date column stored as object	Converted date_added into datetime format
Multiple countries and genres in single cells	Split and analyzed comma-separated entries
Different duration units	Analyzed movie minutes and TV Show seasons separately

10. Conclusion

The Netflix Movies and TV Shows dataset contains 8,807 records and 12 columns. Data cleaning was performed to handle missing values, correct misplaced rating and duration values, and convert the date_added column into datetime format. No duplicate records were found. The analysis showed that Movies represent 69.62% of the dataset, while TV Shows represent 30.38%. The most common release year was 2018, and TV-MA was the most frequent content rating.

The exploratory data analysis identified the United States as the country with the highest number of entries and International Movies as the most frequent genre entry. The average movie duration was approximately 99.56 minutes, while TV Shows had an average of 1.76 seasons. The year 2019 had the highest number of content additions, with 2,016 titles. Ten different visualizations were created to explore distributions, relationships, and patterns in the dataset. Overall, this project provided useful insights into the Netflix content dataset through data cleaning, statistical analysis, and visualization.

11. GitHub Repository

GitHub Repository Link: https://github.com/Its-yusra/Devixo-Solutions-Task-01.git

Project Notebook: Task_01_Data_Analysis.ipynb

12.Google Drive Link

https://drive.google.com/drive/folders/131AAfyI1Xf3T6CuM9hjB77DsntSKbVs7?usp=sharing